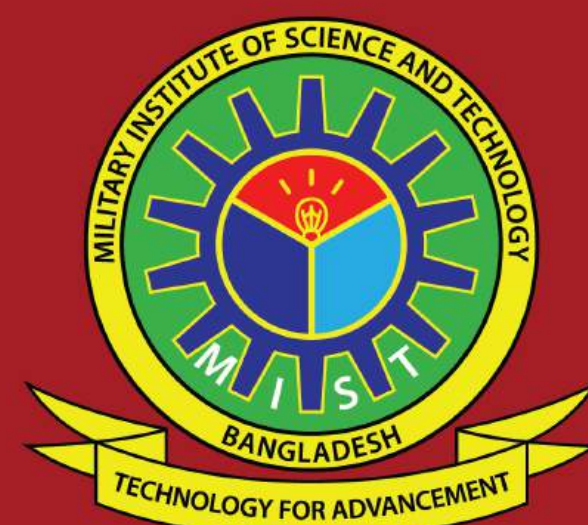
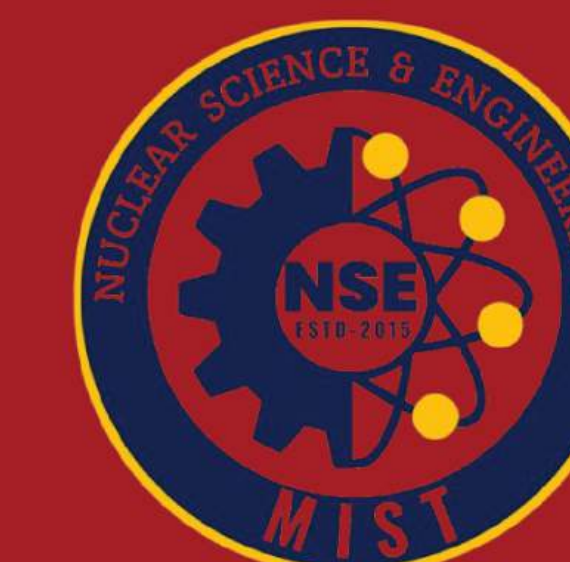




NEUTRONIC EVALUATION OF REFLECTOR MATERIALS FOR THE ALFRED CORE

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Introduction

Given the limited exploration of reflector materials for Lead-cooled Fast Reactors (LFRs), this study compares 4 reflector materials (**Baseline Yttria-Stabilized Zirconia (YSZ), BeO, Graphite, MgO**) for Advanced Lead-cooled Fast Reactor European Demonstrator (ALFRED) core using OpenMC with the ENDF/B-VIII.0 nuclear data library focusing on **optimized neutron economy, uniformity in flux distributions, and increased core lifetime**.

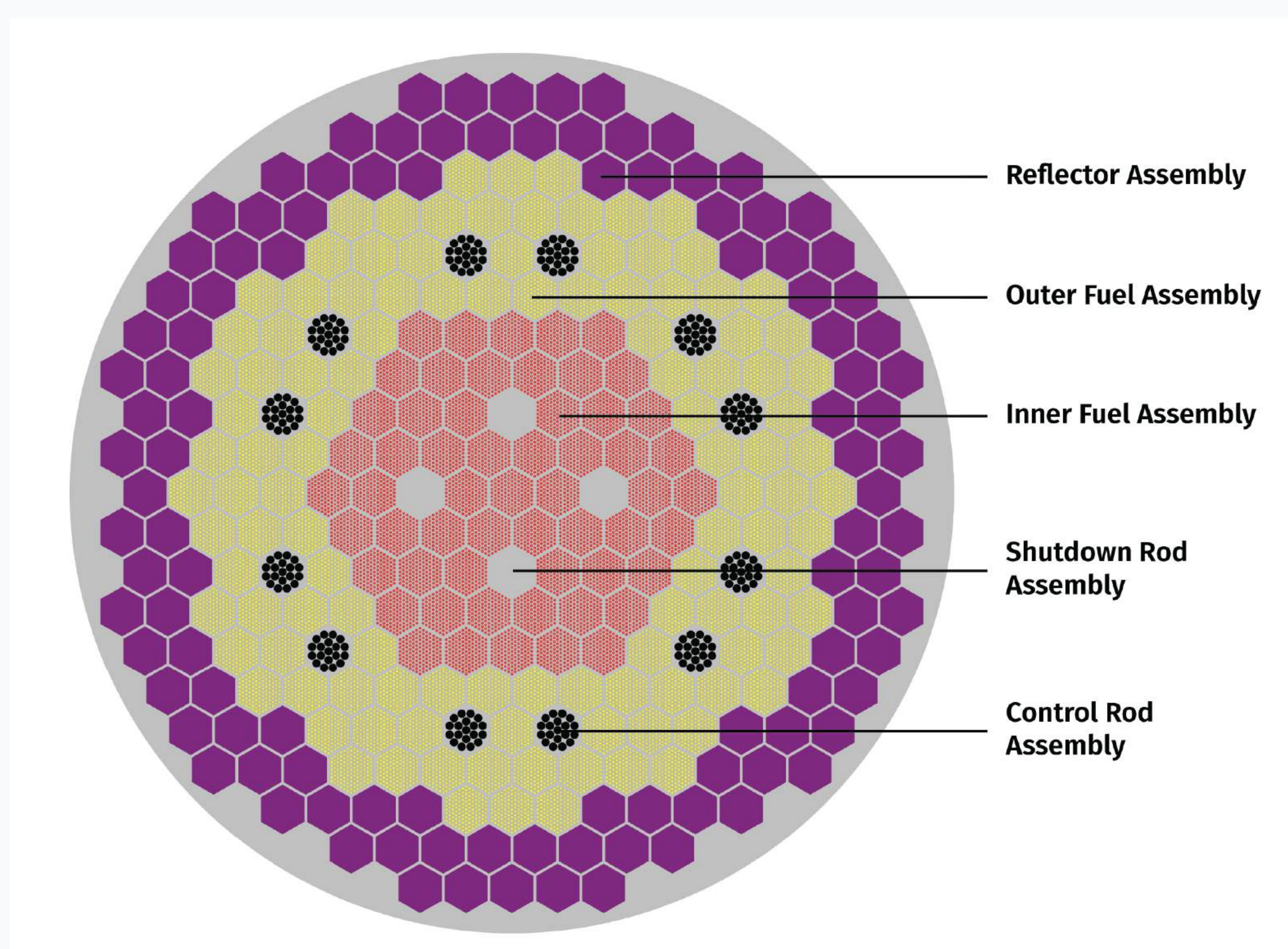


Figure 1: ALFRED Reactor Core

Results and Validation

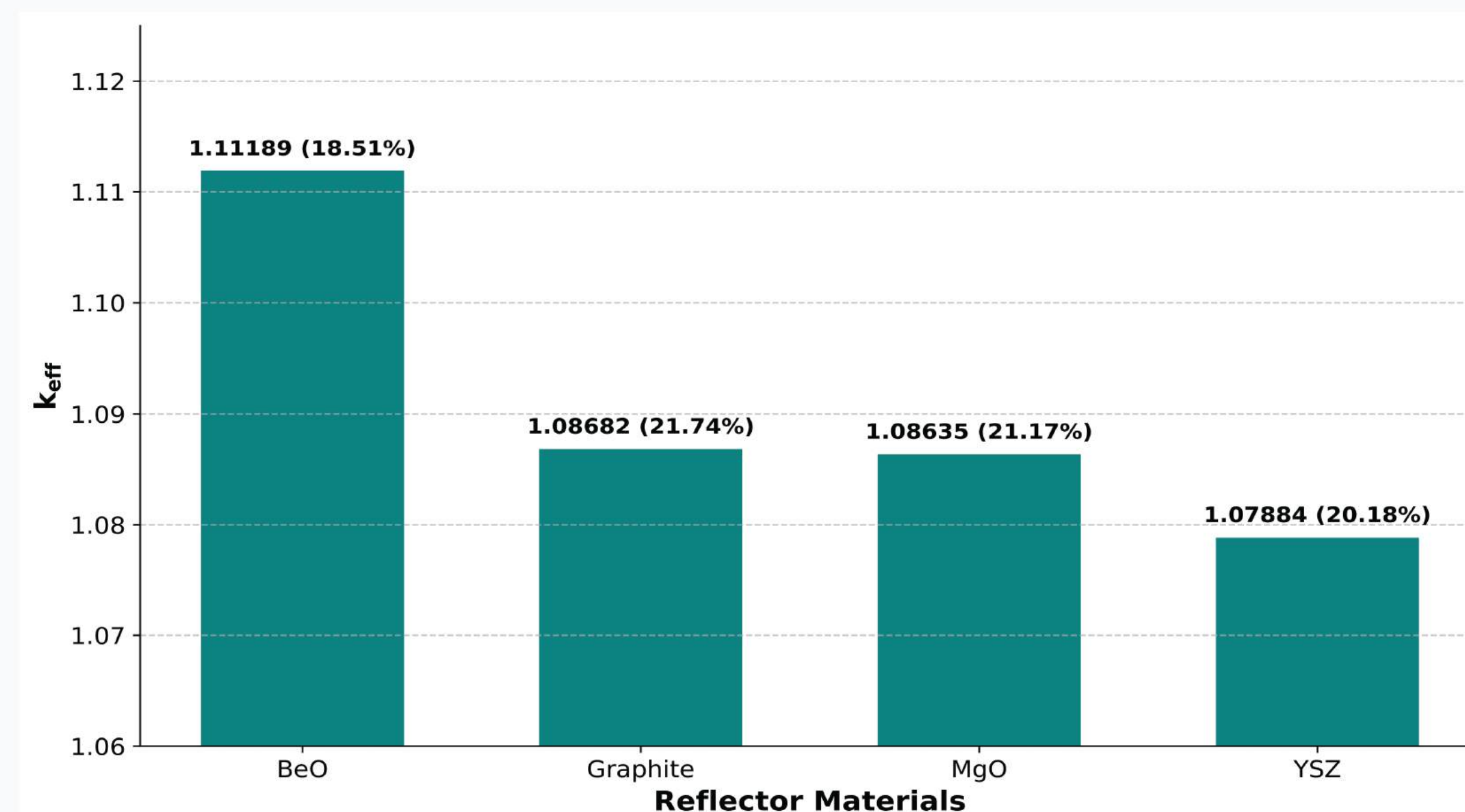


Figure 2: k_{eff} for Different Reflector Materials

The k_{eff} for YSZ matches literature [1] at 1.07767 (error 0.108%), validating the OpenMC model. BeO gave the highest k_{eff} corresponding to the lowest neutron leakage.

[1] Lodi et. al., 2015, Characterization of the new ALFRED core configuration. Report number: ADPFISS-LP2-085 Rev. 0.

Results

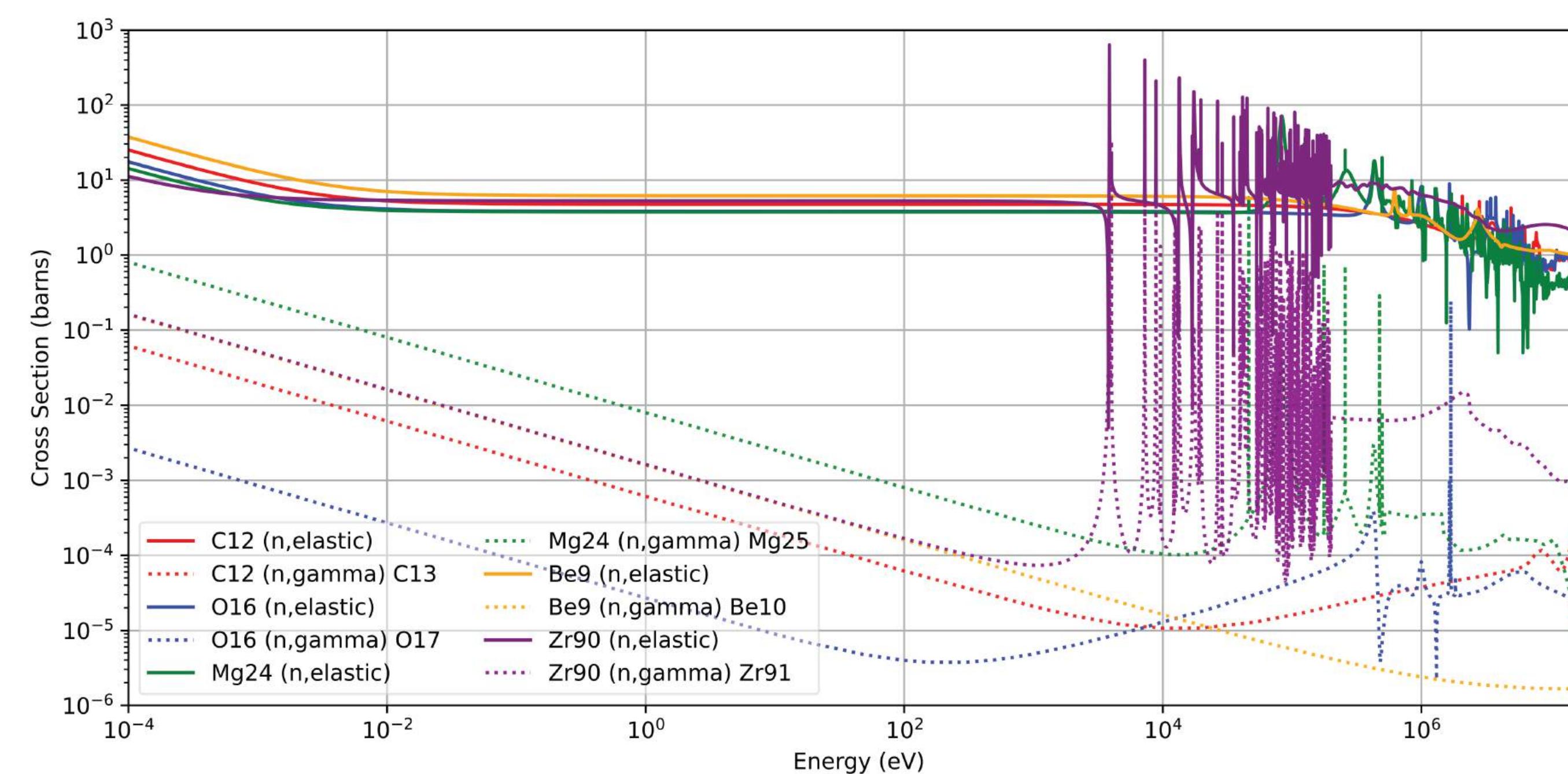


Figure 3: Important Reaction Cross Sections for Significant Nuclides

In fast energy regions, Zr-90 shows prominent elastic peaks, unlike lighter Be-9 or C-12, which give BeO and graphite stronger moderating ability. Be-9 also has the lowest capture cross-section, making **BeO well-suited for preserving neutron economy**.

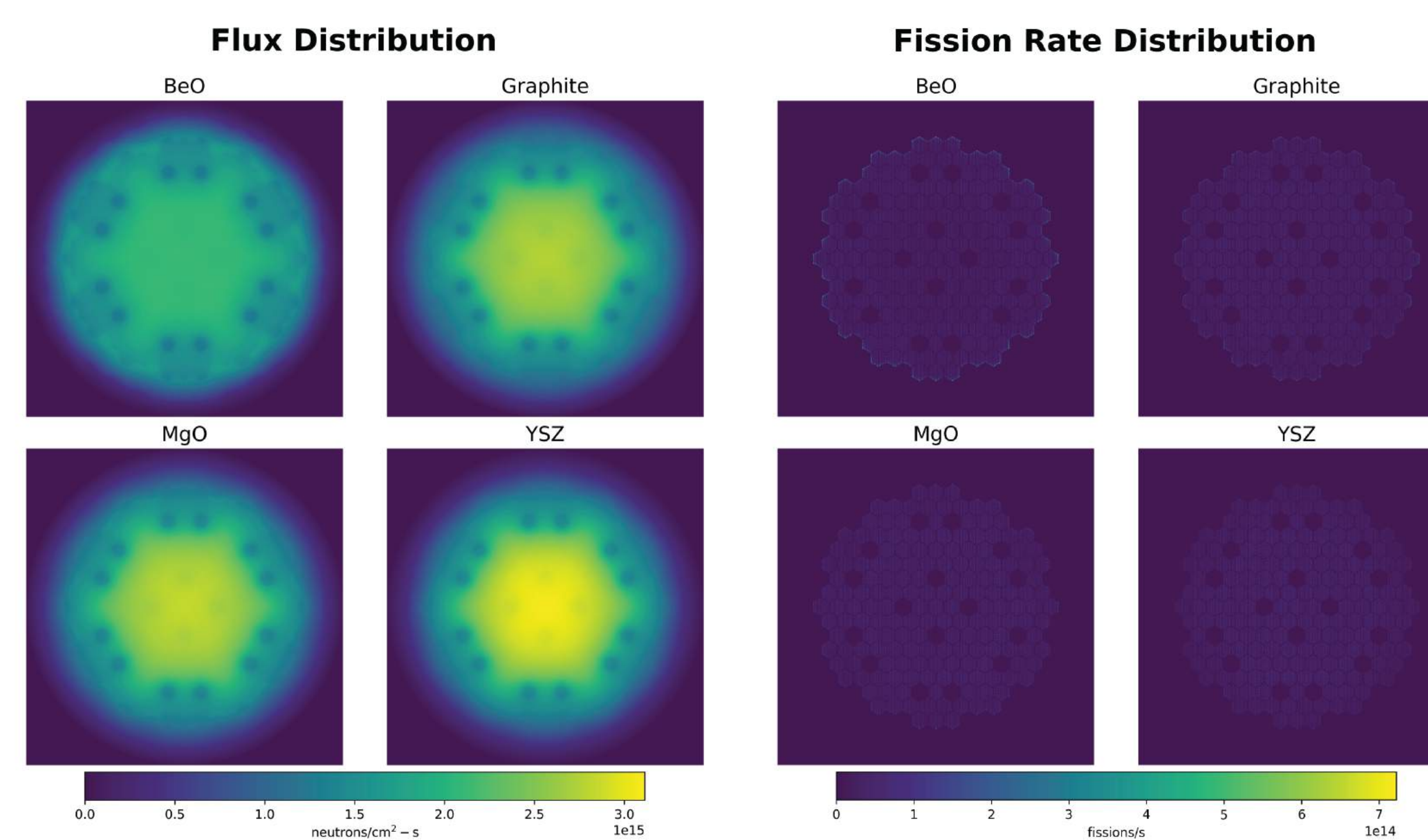


Figure 4: Flux and Fission Rate Distributions

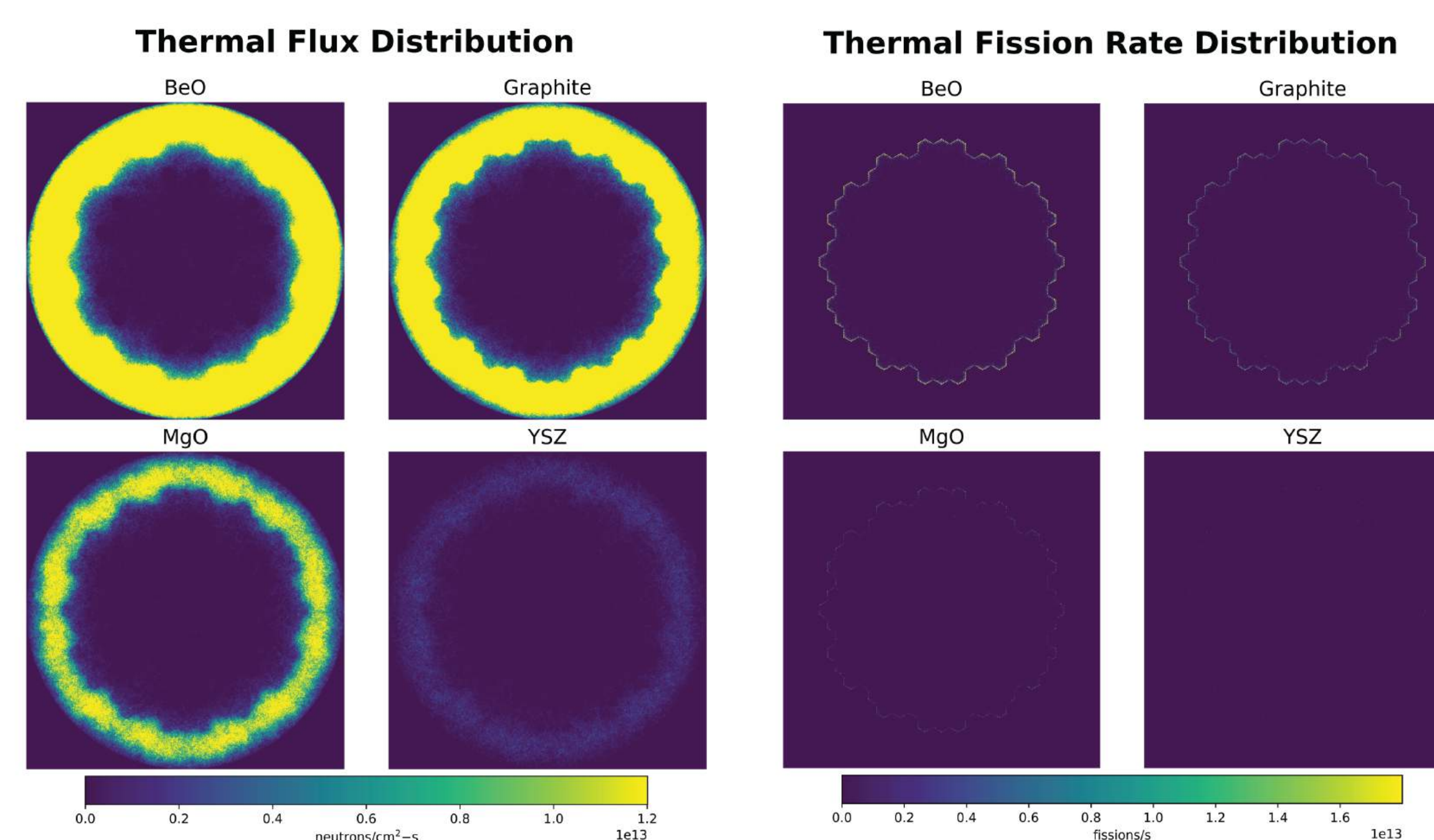


Figure 5: Flux and Fission Rate Distributions with Only Thermal Neutrons

Results

At the outer fuel periphery, fission rates varied with reflector type, with BeO producing the highest peak, followed by graphite. Thermal fission analysis showed that **BeO drives the most thermal fission at the periphery, reaching $\sim 10^{13}$ fissions/s near the core-reflector interface**. This occurs because BeO and graphite act as moderators. The elevated thermal flux near BeO and graphite reflectors explains their higher peripheral thermal fission rates.

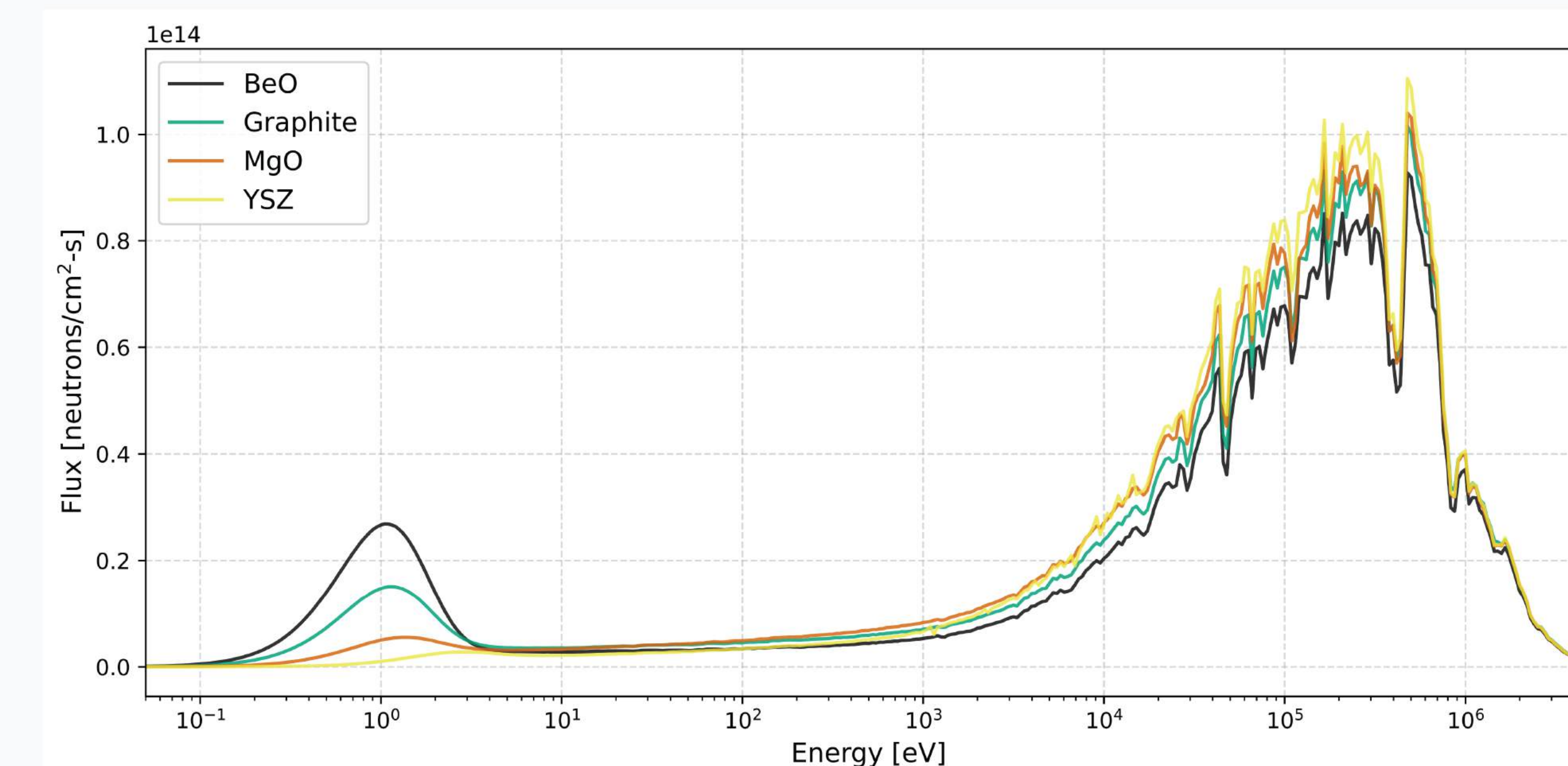


Figure 6: Neutron Spectra

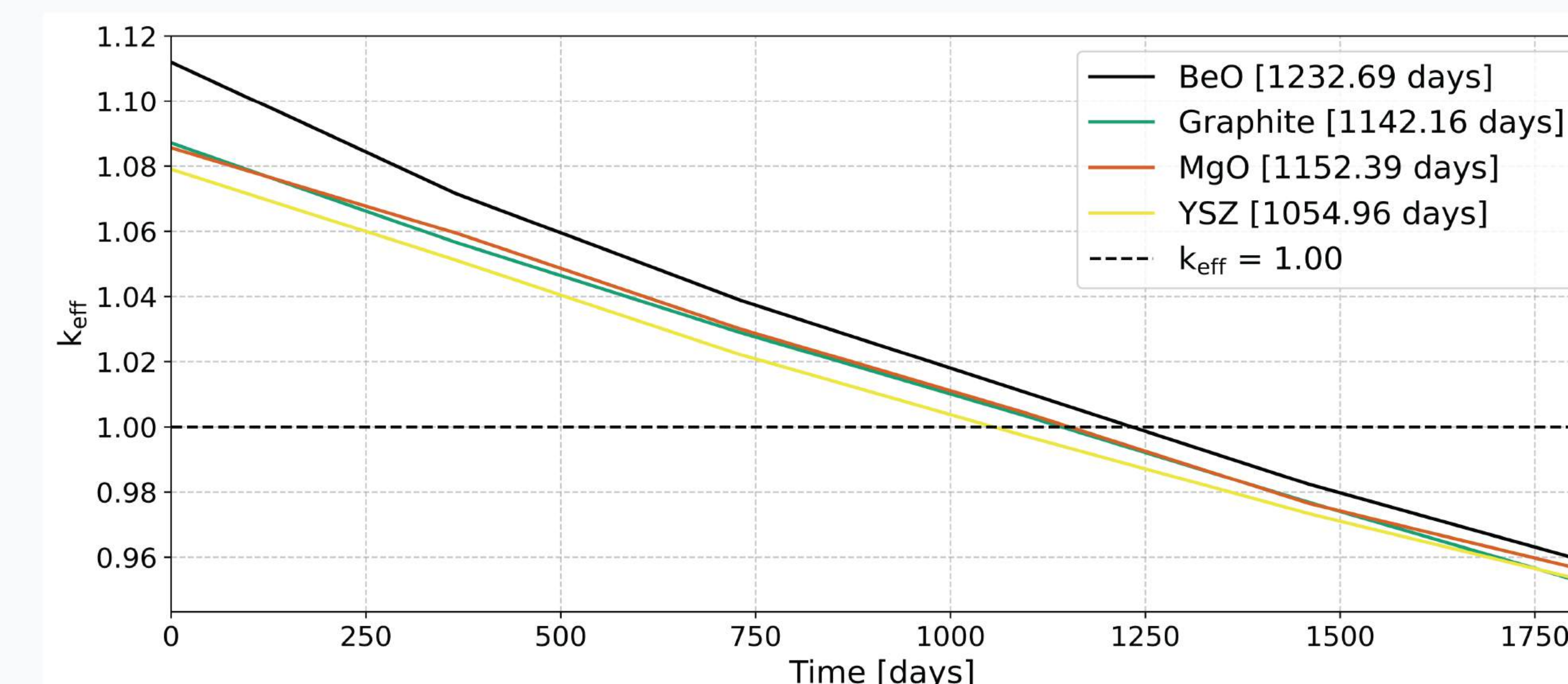


Figure 7: Decrease of k_{eff} with Time, and Time to Reach Criticality

BeO softens the spectrum most, however thermal fission increases too. Overall it gives the **longest core lifetime** among all reflectors.

Conclusion

Based on **cross-sections, k_{eff} , flux uniformity, higher fission rate near the reflector, and increased core lifetime**, BeO proves outstanding as a reflector, despite spectrum softening, and is clearly better suited than Yttria-Stabilized Zirconia from a neutronics standpoint.

Acknowledgement

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